

Justin Hong

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EDUCATION

Massachusetts Institute of Technology

BS in Artificial Intelligence and Decision Making (Course 6-4), GPA: 4.3

Cambridge, MA
Expected May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Construction, Web Design
- **Activities:** MIT Undergraduate Research Opportunities Program, Walker Memorial Basement Radio 88.1FM, MIT Live

TECHNICAL PROJECTS

hwatu -- Browser Engine for AI Coding Agents

2026

- Built a daemon-based WebKit browser in Rust (WebKitGTK 6) that makes visual verification nearly free for coding agents: 13 ms median window spawn (p90 35 ms) from a warm daemon, 216 ms full verify loop
- Designed the agent-facing control plane: JSON-over-Unix-socket IPC protocol, MCP server over stdio, headless-as-window-property rendering, and human hand-off into a tiling window manager
- Sole author across a 3-crate Rust workspace (client, daemon, IPC); 350+ commits with published engineering writeups and benchmarks

Agent-First Linux Desktop

2026

- Composed a development environment where AI agents are first-class citizens: custom browser (hwatu), agent memory servers (MCP), and Wayland compositor tooling orchestrated for autonomous verify-fix loops
- Built and deployed multiple MCP servers (game-economy analytics API wrapper, personal relationship memory) used daily by local and remote agents

Freight Agent -- Subconscious Hackathon

May 2026

- Designed plan-approve-execute UX for an autonomous freight-exception agent on Cloudflare Workers (Subconscious reasoning, Baseten vision/ASR); live SSE trace surfaces draft carrier, customer, and AR messages
- Architected repo partitions (web / worker / db) and built full Next.js storefront with voice-first mic launcher in a 2-hour build sprint

ACL Triage -- Cursor Hackathon

May 2026

- Built Next.js 16 coach dashboard for ACL-injury triage: 60+ athlete rosters, body-silhouette injury mapping, structured check-in flow, and role-based routing
- Integrated FastAPI backend running MediaPipe pose extraction and RAG-grounded rehab retrieval to turn short video + text intake into severity and urgency bands

WORK & LEADERSHIP EXPERIENCE

TrustAI Labs (Y Combinator S26)

Founding Engineer

San Francisco, CA
March 2026 – Present

- Own the end-to-end frontend of an AI product as one of the first engineers: prototyping features and shipping polished, production-ready interfaces
- Built and maintain the full marketing site and interactive product demos used in investor and customer-facing presentations
- Defined the company's visual identity and design system; responsible for all brand, motion, and interaction decisions

ROK Army

Squad Leader & Personnel Administration Specialist

Yeoncheon, South Korea
July 2024 – Jan 2026

- Led a squad in a top-ranked infantry platoon (1st of 48) in CQB room-clearing competition
- Automated high-stakes administrative reporting with Excel macro pipelines, minimizing errors across the battalion

MIT Political History Department

Frontend Developer

Cambridge, MA
December 2023 – February 2024

- Developed a server-side rendered digital exhibition platform with Next.js, improving accessibility and SEO
- Collaborated with MIT History faculty to launch a digital archive of activist murals, increasing visibility to 1,000+ users

SKILLS

- Systems: Rust, Linux/Wayland, Unix IPC, daemon architecture, WebKitGTK
- AI Agent Infrastructure: MCP servers, agent memory systems, LLM tool design, RAG-grounded retrieval
- Web: TypeScript, Next.js, React, Python, FastAPI, Cloudflare Workers, AWS, Vercel

HONORS & AWARDS

- Educational Broadcasting System (EBS) Janghak Quiz Competition Winner (2023)
- Cheongshim International Academy Valedictorian (2023)